

PROJECT SolarAid

Design Proposal

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Executive Summary

Blacktown is a multi-cultural suburb in Western Sydney, which is very vulnerable to climate-related hazards such as floods, extreme heat, and storms. These events make it harder for people to live and put lives at risk. It also weakens and damages the public infrastructures, put heavy strain on energy grids and hampers the daily lives of low-income households and CALD (Culturally and Linguistically Diverse) community. The lack of reliable energy during these crisis leads to the loss of essential services such as refrigeration, lighting, communication, and medical support. To fix this problem this design proposal introduces Project SolarAid — a solar-powered, portable energy generator designed to provide emergency electricity in the times of crisis.

The project starts with a literature review that outlines the vulnerability of the community and the climate related risks its facing. Based on community needs, climate risks and infrastructure vulnerabilities, the team came up with design requirements. The key constraints included maximum energy output of 25kWh/day lasting two weeks, safety around community that follows WHS AS/NZS and NCC electrical standards, durability of withstanding 5,000 N/m² flood pressure and 600°C external heat, environmental performance of emitting ≤ 0.275 kg CO₂-e/kWh, and can adapt against at least three disaster types. The solution also had to be cost-effective, low-maintenance, compact, and rapidly deployable.

Total 16 concept designs were developed and they went through 3 screening phase, among which 5 options fulfilled the constrains and only 1 was chosen. It was a solar-powered generator. The preliminary design integrates a high-energy-density lithium-ion battery, capable of storing 575 kWh enough to supply a standard household for up to 29 days. The generator includes a PV panel array with maximum power point tracking (MPPT) control, a microcontroller (Arduino-based), and a DC/DC converter for efficient voltage regulation. Everything is wrapped around in an Aluminium Composite Panel (ACP) casing with fireproof and impact-resistant layers making it durable against climate related disasters.

The testing has been theoretical and based on mathematical calculations that fulfills design constraints. But it's still is an alpha prototype, so to know how reliable it actually is during times of crisis, physical testings need to be done.

Future work includes making a pilot design, getting funds from local council, running benchmarks in front of qualified professionals. The team aim to make Project Solar aid a long-term solution against various hazards and climate related disasters.

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1 Introduction

The Blacktown community is stricken with extreme weather events, and these events continue to increase each year. On December 9, 2007, a thunderstorm with large hailstones caused severe damage to Blacktown. The storm resulted in 30 injuries and approximately \$415 million in insurance costs. According to Wikipedia, in July 2022, Blacktown was affected by the New South Wales Flooding. Since 2024, Blacktown has experienced 16 earthquakes with magnitudes between 2.0 and 3.5, including three above magnitude 3.0. The local council of Blacktown face the consequences following these national disasters. With the power outages these create, the councils face increasing criticism from the community for not doing enough to keep daily essentials running as well as the hefty costs when repairing or replacing damaged infrastructure, public facilities, and adverse environmental conditions. The issue the group chose to tackle is power for emergencies in the case of extreme events and hazards.

The project's team is making this proposition as the team feel there is a need for societal security during perilous times, which can be solved with a reliable and long-lasting source of energy. This ensures that when in a crisis, homes will remain with power regardless of the state of the electrical grid. By providing a steady and continuous flow of electricity the local community will be able to better cope with the hard times set upon them. By having a reliable source of electricity, the community won't be as vulnerable to the crisis. Without a sure fast solution to this ever-evolving crisis, the community becomes prone to the ever escalating, harsh and untenable adversities, which will result in a rising risk to health, costs of damage and the community's domestic security.

For the project's team to create a steadfast solution to this growing crisis, the team must create a product that is reliable, durable and enduring. Making the product following these specifications the local community can have peace of mind during a crisis that residents will not be left stranded when it comes to essentials for survival.

While the proposed product will serve a similar role as generators, being a backup energy source when the main power source fails, generators are run of fossil fuels which in the current state of the world don't adhere to society's demand for clean energy. Generators run off fossil fuels such as petrol, diesel and other natural gasses, which in the current state of the world, don't adhere to society's demand for clean energy. Generators emit far greater levels of CO₂ than cars which also presents the danger of carbon monoxide poisoning to people in the generator's vicinity. A diesel generator produces 800 to 1500 grams of CO₂ per kWh. This is 1500% to 2900% greater than the estimated carbon output of the proposed product, as the system is charged using solar power and produces 50g of CO₂ per kWh in the beginning of its lifespan before it becomes carbon neutral after three years of use.

In the rest of the design proposal the group investigate the opportunity for a resilient and sustainable energy source for Blacktown's emergency situations.

(Wikipedia contributors 2025) ("Blacktown, New South Wales, Australia, Earthquakes: Latest quakes" n.d.) ("Storm - Sydney, 2007 | Australian Disaster Resilience Knowledge Hub" n.d.) (Kim 2024) ("Generators and carbon monoxide | Power outage Safety | Disasters | SJReady.org" n.d.) (Team 2022) (Katelyn 2023)

2 Literature Review

Introduction

Blacktown situated in Western Sydney is a multicultural suburb that is home to 400,000 people. Around 80% of that population is from a culturally diverse background. According to Census of Population and Housing 2021, Blacktown is one of the fastest growing regions of Sydney. More than half of the population speaks language other than English at home. The Blacktown City Council Social Profile States 2024 states that Blacktown has high unemployment rates, lower income and education compared to other places. It also has high domestic violence, limited access to services for marginalised community including social support and healthy food options. Blacktown is also at a risk due to increase in flooding, earthquakes, extreme heat and other climate related hazards which costs them a lot of insurance money each year. This literature review focuses on impacts, challenges caused by these climate-based disasters and provides solution against them.

Impacts of Water-Based Hazards on the Community

Water-based hazards can affect communities in many ways, including environmental, social, and economic impacts. Environmental impacts can involve damage to both the natural environment and human-made infrastructure. Social impacts include people being displaced from their homes, injuries or loss of life, and disruptions to education. On the economic side, these hazards can prevent people from working, cause significant property damage, and lead to long-term economic challenges for the community.

Natural disasters relating to water alters landscapes, disrupts ecosystems, and contributes to the deterioration of water and soil quality. There is growing evidence, from climate models, that short-term extreme weather events are likely to become increasingly frequent in many parts of the world, threatening the long-term functioning of the eco-system (Bates, Kundell & Winger 2020). Floodwaters can also contaminate soil, and this affects the soil resources that are available to communities. Since many commercial, industrial, residential and agricultural developments have historically been situated adjacent to rivers; they contribute to the contamination of river sediments, and these sediments are often deposited onto the floodplain soils downstream by overbanking river water during a flooding event (Bates, Kundell & Winger 2020). Floods also pose a great risk to not only the natural environment but to man-made structures. Floods expose infrastructure to more risks of structural damage, wearing out, and aging quickly, thus increasing maintenance and replacement costs (Njogu 2021). Flooding in developing countries has a greater effect on their infrastructure than in developed countries due to little planning and development of infrastructure being made by developing countries. Infrastructure vulnerability to floods is a cause for concern in Kenya. In the recent years, Kenya has experienced major episodes of floods leading to high incidents of infrastructure failures and thus socio-economic losses (Njogu 2021). If more extensive research and development was considered by Kenya and its government then possibly these infrastructure failures would decrease thus minimising socio-economic losses.

Water-Based natural disasters have social consequences that are wide ranging and have a great impact on the daily life of communities as well as the safety of its members. The safety of the community is paramount during flood events, and the great news is Deaths that are directly attributable to floods have fallen dramatically over the past century (Graham et al. 2018). These deaths are mainly due to drowning, hypothermia or electrical related and can be defined as direct health costs of a flood. Whereas indirect health costs are extremely hard to measure due to the limited baseline health data and the difficulty in conducting controlled epidemiological studies in a disaster situation (Graham et al. 2018). Indirect costs are also wide ranging so it's hard to measure many qualitative effects like

interruption of health services, environmental impairment, reduced income, displaced populations, and inadequate temporary living conditions (Graham et al. 2018). The impact on daily life for the community affected by devastating flood events can also differ from short term to long term effects. Both long- and short-term effects have a negative psychological impact on members of the community and can result in migration away from the town/city where the flooding occurred, displacement of people is a major problem if it isn't handled properly as it can result in severe economic impact on the area where there are migrating from and to.

The economy is extremely vulnerable during times of natural disaster and can deteriorate due to flooding and can be detrimental to the livelihood of possibly millions of people. Since the early 20th century to the present day, there is a significant increase to the extent of damage caused by natural disasters (Dobrovičová, Dobrovič & Dobrovič 2015), this can mainly be attributed to the population growth of the world as well as more infrastructure being built to accommodate this. In Europe, which many consider a continent not prone to floods, there have been several flood events with material damage in excess of 1 billion EUR (Dobrovičová, Dobrovič & Dobrovič 2015), this is also not counting the extensive non-material damage that has occurred in a place which experiences a lot less events than countries in south-east Asia. Direct material damage can in turn affect the output of businesses, if material damage is large, the loss in overall output will generally exceed the loss of income in damaged businesses (Davies 2015). This can have a devastating effect on businesses and in most cases cause them to close, the economic output of a community means higher unemployment and less income to the community. This can take years for a community or region to fully financially recover from a one-off flood event.

Effects of extreme heat on Community

Extreme heat happens when the environment is exposed to excessively hot weather and sometimes followed by very high level of humidity. During the events of extreme heat the buildings, roads and concretes retain heat longer than usual and because of that it feels way hotter in those times. According to Bureau of Meteorology, In Australia, it's typically defined by maximum temperatures above 35°C, sustained over several days. It has claimed more lives than any other natural hazard in Australia.

In times of extreme heat it's always the children and elderly people that suffers the most. According to volume 67 of International Journal of Disaster Risk Reduction, from 2000 – 2018 at least 354 people have been killed by heat waves. During heatwaves people suffer from heat related illness and they include – Heat Stroke, Heat Exhaustion, rhabdomyolysis, heat syncope, heat cramps, and heat rash (CDC – Heat related illness – September 10, 2024). These heat related illness is very severe for elderly, infants and children, people with disabilities and people living without any cooling systems.

During heatwaves the high temperature also put a lot of strain on the electricity grids which increases the risk of power outage. The strain on the power grid happens due to high usage of air conditioners and coolers. The case study done by Bubathi et al. 2023 found that in Western Sydney, including suburbs like Blacktown faces as much as 10°C higher temperature than the coastal areas. As the temperature starts to climb residents starts to rely on air cooling systems which cause a massive spike in electricity demands and puts a heavy strain on energy grids. Because of the strain put on the energy grids, the likelihood of power outage increases which sometimes can result in a massive blackout.

Backup power

Flooding and extreme heat aren't the only disaster affecting the communities in Blacktown. There are others such as severe storms, bushfire smoke, and strong winds. Blacktown is not situated in a direct bushfire zone but it experiences smokes and haze from the nearby bushfire zones, which causes bad overall air quality. Strong winds and storms can damage power infrastructure and lead to extended outages, sometimes massive blackouts (Johns, R & Parliamentary Research Service 2023). The lack of reliable energy during crisis leads to the loss of essential services such as refrigeration, lighting, communication, and medical support. Because of these increasing weather events, having a reliable backup power source is a must. Blacktown has a multi-cultural community and its vulnerability to climate makes it crucial to establish backup energy to both households and community level. So installing private systems, community hubs should offer the public a sense of comfort during the times of disaster. In addition, during hazards backup power can promise the public of public health and safety and social stability. Ultimately installing backup systems in communities will empower them to be self sufficient during the times of crisis.

Problem CALD communities face during disaster

Culturally and Linguistically Diverse (CALD) communities are often at a disadvantage due to language barriers, lack of recognition of overseas qualifications, cultural misunderstandings and discrimination. These obstacles are heightened during times of disaster and can have a much more severe impact on these types of communities compared to less diverse communities. Australia compared to other countries is extremely diverse, the population comes from over 200 different countries, with around 300 different languages spoken at home (Ogie et al. 2018). Migration to countries like Australia is increasing and at the same time climate change is posing greater risks causing more natural disasters and devastating events. Migrants and CALD communities are particularly vulnerable to natural disasters, including the previously mentioned barriers that these communities face but more specifically to adverse events they face unfamiliarity with new environment, lack of support networks, inexperience with disaster events, lack of emergency plans, limited literacy and fluency in the dominant language inability to access community resources due to physical and social isolation, low income and poor socio-economic conditions to support recovery from disasters, and limited insurance to minimize loss (Ogie et al. 2018).

Conclusion

Blacktown is always at the risk of weather and climate related disasters and they are increasing because of climate change. Because of floods, heatwaves, bushfires and many other climate related events put a heavy burden on the population, public health, economy, public and emergency services and local infrastructure. The suburb has very unique and multicultural community with more than 80% people are from culturally and linguistically diverse backgrounds. Because of that the community has to face social and economic challenges, it has a high unemployment rate, low education levels and limited access to services. On top of that these challenges are made worsen due to these climate related disasters. Water based hazards such as floods damage the infrastructure making them weak and damaged which increases the replacement and maintenance costs. Extreme heat puts a heavy strain on the energy grids causing power spikes and power outages. There are also bush fires, storms, heavy wind and they often result in death. To build long-term resistance, it is important to invest on sustainable and reinforced infrastructure, stronger power grids and reliable source of backup power. With proper planning and help from the community it will be easier to minimize the risks and protect the residents.

3 Design Requirements

3.1 Opportunity and Problem Statement

For the project, it is driven to boost the community's resilience when facing an increase in the amount and intense extreme weather situations. The team is aiming to provide an adaptable and reliable power source that sustains the community's needs during any extreme weather conditions they may face.

How might families and households in Blacktown reduce the lack of reliable power caused by extreme weather conditions, as this affects the community's way of everyday life, by being unable to sustain fresh food and drinkable water.

3.2 Design Requirements

Constraints

- The design must be efficient and generate minimal energy waste. During times of disaster, the community will need a design that can provide for many people with maximum output. 25 kW per day is needed to survive 2 whole weeks
- The design must be safe for the community. During times of disaster, the community will not need to worry about getting injured from an unstable design when they are panicking. For this, it needs to follow one of the given guidelines:

- i. Work Health and Safety Regulation 2017

<https://legislation.nsw.gov.au/view/html/inforce/current/sl-2017-0404#ch.4-pt.4.7>

- ii. WHS Act 2011

<https://legislation.nsw.gov.au/view/pdf/asmade/act-2011-10>

- iii. AS/NZS 3000

<https://www.powerandcables.com/wp-content/uploads/2019/07/AS-NZ-3000-2018-Australian-New-Zealand-Wiring-Rules.pdf>

- iv. NCC Volume 2

<https://ncc.abcb.gov.au/system/files/ncc/ncc2022-volume-two.pdf>

- The design must be made from durable materials. When in a time of disaster, the design must withstand any damage from the surrounding environment and still be functional as usual. With that in mind, for the design want the materials need to withstand 5,000N/This is because this is the minimum force produced by flooding in Blacktown and must withstand 600 degrees. This is because this is the average temperature of the outer part of the fire when in a bushfire.
- In the design, the group are aiming to make it environmentally friendly. This is because creating a product that doesn't produce much greenhouse gas will help slow down the climate impact. In 2024, New South Wales produced 0.68 kg CO₂-e/kWh; in the future, they want to reduce it by 43%. Therefore, for the design, it cannot produce more than 0.275 kg CO₂-e/kWh.

- For the design, want it to be adaptable for all situations. With this in mind, the design will be more suitable to help the community during a wide range of disasters, instead of a select few. With that in mind, the design has to be adaptable for at least 3 different extreme weather conditions.

Objectives

- The timeline for this design will need to be up and running within 2 years. Blacktown requires an immediate response to its changing weather patterns and increasing frequency of extreme weather events and hazards.
- For the design, would aim for low maintenance and easy repairs. This will make it easier for Blacktown to focus on improving their community and only need to worry about the design.
- In the design, the team is aiming to make it easy to access to reach all areas of the design. This will make it easier for the maintenance to be done quickly if they can reach all parts and be more sustainable.
- With the design, the team wants it to be cost-effective. This is because if the design is low in cost, it will be able to produce more products for the community to face any crisis.
- The team wants the design to be space efficient. This is because it is user-friendly, so it will be easier to transport, store, or use in tight spaces. Also, the team wants the design to be energy efficient and compact.

4 Methodology

This design process followed a structured approach that is based on the conceptual and preliminary design phases; after fulfilling these phases, it will produce a suitable design to solve the problem.

4.1 Conceptual Design Methods

The conceptual design phase involves generating a diverse range of innovative ideas to solve the key challenge that is: ensuring that Black Town has reliable energy during severe weather emergencies. To find the most optimal design, the group will follow a three-stage process:

1. Brainstorming and Idea Generation

The first stage is developing a wide range of ideas without any limitations or judgment. By doing this, it helps the group members to think creatively, ranging from a simple and straightforward approach to an unusual approach by thinking outside the box.

Some techniques that will be used in this stage:

- Mind mapping is where the group visualises any connections between the ideas. By using this, it helps the team to connect different ideas. By doing this, it can benefit the design by seeing that different ideas interact (for example, combining a solar panel and a portable generator) and can be a better idea than just one idea.

- Reverse brainstorming, this is where the team creates some ideas on how to make the energy problem worse; this sometimes can create an unexpected solution to the energy problem. By using this, it can help the team by focusing on ideas to make the problem worse; by doing this, it can lead to an unexpected solution. This can benefit the design by discovering vulnerability in the system, and it can lead to an unusual solution (for example bad idea: make the powerline more fragile so it can easily be destroyed by disasters. The Solution is to strengthen the powerlines)
- Inspiration from existing solutions, this is where the group look at how other companies or communities have solved a similar energy problem. By using this, the team will be able to avoid designs that may seem good on paper but in the real world may not. This can benefit the design by saving time and resources by seeing proven solutions, but it can also benefit the team by avoiding mistakes that others have made trying to solve a similar problem.

The goal with these techniques is to maximise the creativity of the initial idea generation.

2. Primary Screening: Checking against design constraints

Through the next stage, all the generated ideas will undergo a primary screening based on the design constraints (for example, durability of the materials, environmentally friendly, adaptable for any situation and safe around the community). Each idea will be assessed by using a binary (yes or no) system:

- Yes, is where the idea meets the criteria for all design constraints, and it is feasible to be implemented into the real world.
- No, is where the idea failed to fulfil the constraints requirement, and this will eliminate the idea from moving to the next stage

Once all the ideas have gone through each constraint, it will minimise which ideas are viable for the energy problem and which ideas will be removed due to being unworkable for this situation.

3. Secondary screening: Scoring against the design objectives

The remaining ideas from the primary screening will advance through to secondary screening, where they will be assessed against the design objectives (for example cost cost-effective, time-efficient, low maintenance, sustainability and size effectiveness). To ensure that the most optimal design will go through, the group will use a traffic light scoring system, this is where:

- Green, exceed the design objective and give the idea 3 points
- Yellow, this is where it just meets the design objective and will give the idea 2 points
- Red, this is where the idea falls short of the design objective and will give the idea 1 point

Each idea will be scored by each objective, and the points will be tallied up after each idea has been scored by each objective. By using this method, it will:

- Minimises bias by using a metric system.
- Shows trade-offs (for example, an idea could score high on low maintenance, however, it could be low on affordability).

- Identifies which idea is more suitable for this energy problem.

By the end of the three stages, the team has found the highest potential design for Black Town's energy reliability problem. This top idea will advance to the Preliminary Design stage, where it will undergo real-world testing.

4.2 Preliminary Design Methods

The preliminary design phase is focused on developing and improving the preferred design idea that passed through the conceptual design process into a functional, tested and viable solution. To create the most optimal design, the group will need to go through three phases, which are embodiment, testing and verification. Each one ensures that the design meets all the requirements before the group gets the final design.

1. Embodiment

For this phase, the objective is to transform the preferred design into a physical and measurable design by defining the structure and what components that is needed.

- First step for this phase, the team would need to find what components are going to be used for the design. Specifically, the group would need the exact sizes, materials and arrangements for all parts (for example, the capacity of the battery, the frame of the design and the solar panel dimensions).
- For the next step, the team need to make some technical drawings and a layout of the design. By creating a 3D model drawing it would help visualise the design and also see how the team would be able to assemble all the parts and see how it fits. Also, with the design, the group could create a 2D schematic demonstrating how the system will be organised.
- Lastly, the next step, the team will need to find the most suitable materials for the design. One way to find the most suitable is to compare materials with one another (for example, the durability, weight or cost).

After doing all these steps, the outcome will be a detailed design ready for testing.

2. Testing

For this phase, the objective is to validate that the design will be functional against the real-world conditions.

- One method to see if it is validated is to use calculations, by using engineering equations, the team would be able to predict the behaviour of the system. This is helpful because using calculations will reduce the costs and time for testing the physical design, therefore, the team could put more time into developing the other parts of the design.
- Another method is to create a digital or physical prototype; this is where the team build a scale model (for example, a 1:10 prototype of the exterior design). This will help the team to test if the design is functional and be able to be usable during extreme weather when there are power outages.
- Another method is doing an environmental test, which is where the group tests if the design will be able to withstand extreme weather (for example, by doing a temperature test on the material to see if it would be able to withstand a bushfire). This will be helpful because the

team will be able to catch a failure in the design early and be able to change or improve that area.

By doing testing, the outcome will demonstrate whether the design will be functional for the real world, or it will expose how flawed it is.

3. Verification

The final phase of the preliminary design is verification; this is where the final design will be assessed by the Statement of Compliance. The Statement of Compliance is where the final design will be assessed against design constraints and see if the successfully passes the requirements.

The reason the group would use these phases is to bridge the gap between an idea to a real design. This ensures that the design is functional, safe and ready to be implemented in the real world.

5 Results

5.1 Conceptual Design

In the ideation phase, the team have created a diverse range of ideas:

- Option 1: Portable Generators/Battery Units: These are mobile power storage units employable anywhere they are needed. This is useful for any emergencies, temporary power outage or a remote location.
- Option 2: Have a storage with spare energy from a generator: This is installing an energy storage system with an existing generator. This is useful to that there is case the generator fails.
- Option 3: Increase the durability of the power poles and lines: By increasing the durability of the power lines to withstand extreme weather. It will reduce the outages caused by extreme weather.
- Option 4: Split up the power grid: Dividing the grids into smaller areas. This will be useful if one location has a power outage, the rest of the power grids will still be operational.
- Option 5: Underground power lines: This involves burying cables underground to protect them from some extreme weather. This can be useful to reduce weather-related power outages.
- Option 6: sensors on generators when emergencies: This is installing smart sensors on a generator to show if something is overheating, malfunctioning or rising water levels. This is useful to alert the maintenance before the generator is going to breaks down.
- Option 7: vehicles that can transport spare energy: This is mobile battery units that deliver emergency power to locations in disasters. This is useful for responding when power grids are down due to emergencies.

- Option 8: Floodwalls: A barrier to protect power stations from flooding. This is useful in preventing water from damaging the electrical system.
- Option 9: Improve drainage: This is improving the stormwater system to prevent flooding. This will reduce the water damage to power systems by reducing the water level.
- Option 10: Improve building durability: This is improving the building durability to withstand extreme disasters. This ensures that the building is able to use energy during these extreme weather conditions.
- Option 11: Solar power: This is installing solar panels that provide renewable energy. This is useful by reducing the reliance on the power grid; by using this, it can still provide energy during power outages.
- Option 12: Fuel cell: This is where they use hydrogen to generate electricity with high efficiency. This is useful for reducing fossil fuel use, but also as a good backup power source.
- Option 13: Geothermal energy: This uses the heat from the earth to generate a renewable energy source. This is useful because it is not weather-dependent and is good for energy resilience for a long time.
- Option 14: Nuclear energy: This provides huge, reliable, low-carbon power by using radiation. This is useful because it is highly resilient if it is properly secured.
- Option 15: Hydro generators in the waterways: This is a system is hydropower inside a river and is used for local energy generation. This is useful because this works when the water flow is continuous.
- Option 16: Energy efficient Shelter home: This is a designed building that uses less energy and with better insulation. This is useful due to reducing the amount of energy the building therefore, when in a disaster, it won't put too much strain on the power supply.

The next step in the conceptual design was going through the primary screening stage, where the team eliminated half the ideas by using the design constraints (shown below in Table 1). The design constraints used were energy efficiency, safe around the community, environment friendly, adaptable and is the design be durable. The ideas that fulfilled the design constraints were, Option 1: Portable Generators almost didn't go through to the next stage because it produces more than 0.275 kg CO₂-e/kWh however if the design used solar panels it would produce less carbon emission at 41g CO₂-e/kWh therefore it will pass it will get through the next screening, Option 6: Sensors for emergencies passed through because the sensors can quickly detect emergencies, also helps community to be safer and modern sensors are durable to survive in harsh weathers, Option 9: Improve drainage systems this passed through because in reduces the impact from flooding and it is designed to last for years under

different weather conditions, Option 15: Hydro generators this went through because it use flowing water to generate clean energy and can also operate during different weather conditions and Option 16: Energy efficient shelter home because this pass through reduces the energy usage and provides the safe area for the community during extreme weathers.

	Portable generators	Increase availability of power grids	Split up power grids	Underground power lines	Sensors on generators for emergencies	Vehicles to transport energy	Floodwalls	Increased drainage systems	Improve durability of building	Solar panels	Fuel cells	Geothermal energy	Nuclear power	Hydro generators	Energy efficient shelter homes
Efficient	✓				✓	✓	✓	✓			✓	✓	✓	✓	✓
Durable	✓	✓		✓	✓	✓	✓	✓	✓				✓	✓	✓
Safe around community	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓
Adaptability	✓	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓
Environmental Impact		✓			✓		✓	✓		✓	✓			✓	✓

Table 1: Primary Screening

Lastly in the conceptual design is the secondary screening stage (shown below in table 2), it indicated that Option 1: Portable Generators with solar panels is the preferred design due being adaptable for any situation that Black Town may face, and it is one of the best ideas that can maintain its productivity and lastly it has a good balance in cost-effectiveness and time efficiency when producing and fixing. A close second was the sensors; it was cheaper to produce compared to the generator, however, it lacks in the of how long it needs to be produced, this is because the sensors need to be coded for Blacktown conditions compared to the generator. Lastly, the sensors take more effort to fix a problem in the system compared to the generator.

	Cost Effective	Time Efficient (Time To Produce)	Low Maintenance	Sustainability (Easy To Fix)	Size Effectiveness	Total
Portable Generator W/Solar Panels	2 pts	3 pts	3 pts	3 pts	3 pts	14 pts
Sensors For Emergencies	3 pts	2 pts	3 pts	2 pts	3 pts	13 pts
Hydro Generator	1 pts	1 pts	1 pts	1 pts	2 pts	6 pts
Energy Efficeint Shelter Home	2 pts	2 pts	3 pts	3 pts	2 pts	12 pts

Table 2: Secondary Screening

5.2 Preliminary Design

Design Process

“At its most basic, charging a battery through solar panels is done by connecting a photovoltaic panel to a battery and using a diode to prevent the reverse current from the battery. This basic design allows for the product to be low in cost. However, some errors that occur can include the panel can only charge the battery if its voltage output is greater than the battery’s voltage, when the difference in the voltage between the panel and the battery decreases fewer current flows into the battery, and it is not always working at an optimal rate to convert solar into electricity.

The drawbacks from the above issues listed can be improved by adding a controller between the panel and battery. A controller can be a DC/DC converter which should be able to increase and decrease the voltage output of the panel. In addition to the DC/DC converter, a program can be used to optimize the solar power’s conversion into electricity.

For the PV panel to optimise the average efficiency when producing energy, cooling must be used as the difference between standard testing conditions are real world conditions without cooling is 21% to 10.5%.

As there is a relationship between the voltage output of solar panels and the voltage intake of the battery, the prototype must find a balance between the two that optimises its charging.

For the algorithm, to get the most available solar power from the panels it should; continuously track the voltage of both the solar panel and the battery, adjust the power flow using a buck/booster converter, maximise power extraction using Maximum Power Point Tracking (MPPT), and be able to adapt to environmental changes. The MPPT is used to find the ideal operating point for the solar panels output curve. The algorithm can be utilised in the circuit by connecting a microcontroller such as an Arduino.

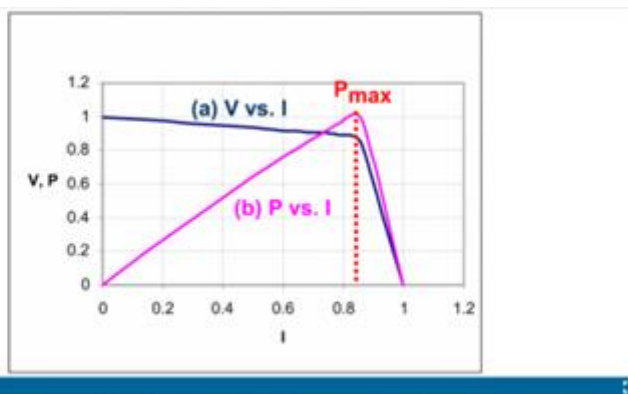
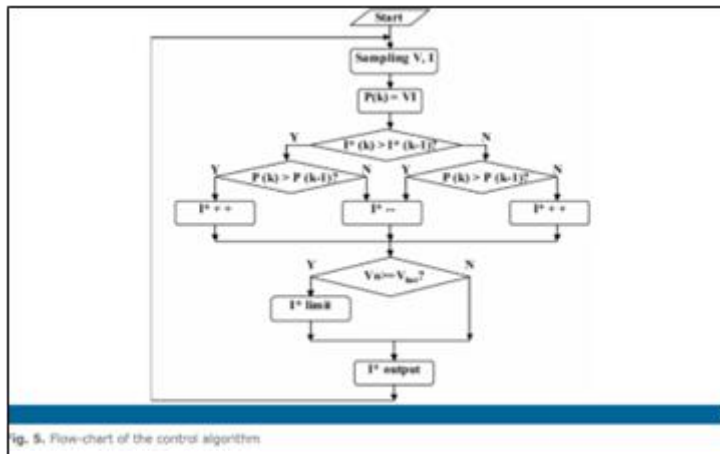
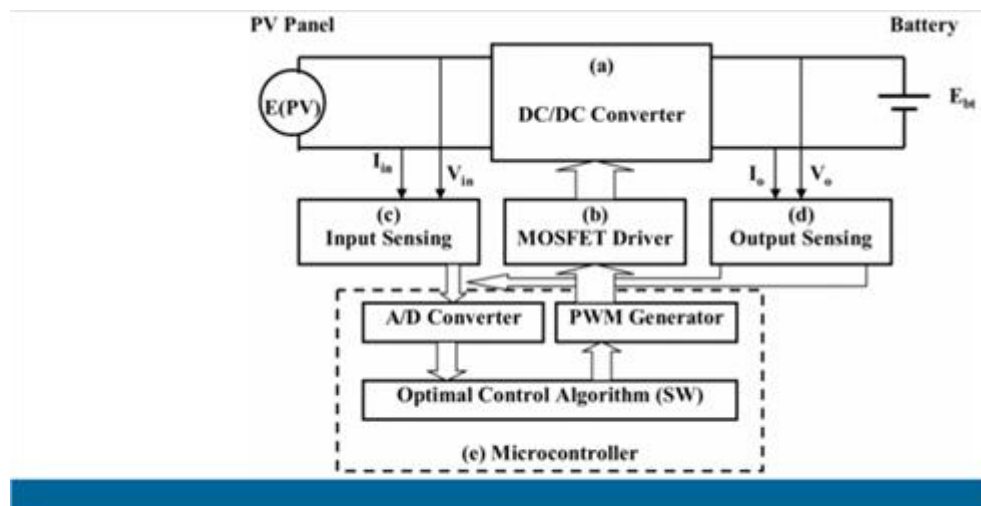


Fig. 3. Typical output characteristics of a PV Panel Y axis: Voltage (V) and power (P) in R.U., X axis: current (I) in R.U.



To make an optimal controller it includes the components; DC/DC Converter, MOSFET Driver, Input Sensing Circuit, Output Sensing Circuit and, Microcontroller.



This diagram of an optimal controller outlines how the DC/DC converter is used to implement the bucks and boosts of the voltage which has been calculated in the microcontroller using the information gained from the components above. This calculation us done by the algorithm previously mentioned.” (“Design of a solar powered battery charger” 2009) (Akrouch et al. 2024)

“The ability of a battery to intake power from the solar panel is affected by its state of charge and internal resistance. When the battery is fuller, it is more likely that the charging current is too high, the terminal voltage rises quickly which triggers the controller to limit or stop the inflow of electricity. This means that while there may be a surplus of inflow to the solar panels, the battery may not accept all of the energy which will cause waste. This is a far greater issue to the efficiency of the transfer of energy with fixed voltage setpoints which are found with on-off type charging controllers, where an algorithm isn’t involved.

Potential limitations:

With smaller battery systems with high charge rates such as the C/20, they often stop charging earlier, which reduces energy storage and load availability.

Larger battery banks are more capable of absorbing high amounts of currents running from the solar panels, such as in times like the middle of the day, which improves energy use and performance at night, when there is no more sunlight.

When charging late in the day when there is less currents inflow from the solar panel, it outlined that the behaviour of battery charging isn't linear.

To best optimise the system the following should be considered:

- Matching the array size to the battery capacity. By oversizing the array without increasing the battery size can reduce the actual amount stored.
- Use charge controllers that are capable of adapting to battery conditions rather than constant types.
- Avoid charging at a current that is too high (aim for C/25 to C/50) to reduce early cutoffs and improve the longevity of the battery.
- Adjust charge controller voltage setpoints for a better compatibility with battery acceptance rates.”

(“Field investigation of the relationship between battery size and PV system performance” 1993)

Requirements for power to be held by battery

“As Blacktown is located in the Greater Sydney area, the team can use the average annual daily electricity usage to estimate the amount of electricity the battery has to produce. The average electricity usage is reported as 19.2 kWh per day” (Fan, MacGill & Sproul 2015)

Battery Power

“A light vehicle battery cell such as what is used in scooters have a capacity of 7 Ah and nominal voltage of 3.7V given 25.9 Wh per energy cell and has a max watt output of 129.5 W.

A light vehicle module such as with used in a full scooter pack, with a module configuration of 14 Ah – 26 V, which is two cells parallel and seven in series produces 364Wh or 0.364 kWh per module.

A HEV Gen 2 cell, such as what is used in high power HEVs, has a capacity of 5.5 Ah, a voltage of 3.6 V and produces 19.8 Wh of energy and has a max watt output of 900 W.

To find the number of each battery type is required to provide a suitable amount of electricity for the hospital use the equation:

$$\text{Number of Unit Required} = \frac{\text{Required power per hour (kWh)}}{\text{Energy per unit per hour (kWh)}}$$

Light vehicle battery:

$$\text{Number of Unit Required} = \frac{3589.943 \text{ (kWh)}}{0.0259 \text{ (kWh)}}$$

$$138608 \approx 138607.84 = \frac{3589.943 \text{ (kWh)}}{0.364 \text{ (kWh)}}$$

Light vehicle module:

$$\text{Number of Unit Required} = \frac{3589.943 \text{ (kWh)}}{0.364 \text{ (kWh)}}$$

$$9863 \approx 9862.48 = \frac{3589.943 \text{ (kWh)}}{0.364 \text{ (kWh)}}$$

HEV Gen 2 cell:

$$\text{Number of Unit Required} = \frac{3589.943 \text{ (kWh)}}{0.0198 \text{ (kWh)}}$$

$$181311 \approx 181310.25 = \frac{3589.943 \text{ (kWh)}}{0.0198 \text{ (kWh)}}$$

These calculations are assuming perfect conditions and a 100% depth of discharge. So as a buffer for this it should be considered an extra 10-20% more cells to make up for the loss of energy.

This large number of cells required to run the hospital for an hour isn't feasible in reality. While it may work for smaller essential services such as food storage, for a more energy demanding service, a battery with greater storage must be used.

As these PV panel powered generators are also designed to serve the purpose of powering the storage of essential produce and emergency services, the smaller batteries listed above are more feasible to use. However, because the goal is for them to be able to adequately service all of the targeted areas, it will be better to use a battery with greater storage capabilities." (Horiba et al. 2005)

"An option for the type of battery used is a lithium-ion battery (li-ion battery), having the greatest volumetric energy density, which is the amount of energy that can be stored in a battery relative to its size. This means using these types of batteries the group will be able to achieve the greatest amount of energy storage for minimal battery size.

Sulfur based batteries while having a very high theoretical energy density of 2772 Wh L⁻¹ are limited to issues a poor life cycle, lower voltage output and the polysulfide shuttle effect, which is where Sulfur compounds drift over to the anode and undergo unwanted chemical reduction, causing the battery to slowly discharge and lose capacity. As the product being developed is to be used in unplanned crisis situations, it isn't optimal to have the unused battery slowly losing its capability to function as intended and is therefore less reliable during these times.

The type of battery with the most optimal specifications is a lit-ion battery with a high nickel cathode and silicon anode. This has an energy density up to 1150 Wh L⁻¹. The Amprius developed silicon-anode lithium-ion battery is the current best battery to use for the greatest volumetric energy density with a value of 1150 Wh L⁻¹.

From the formula

The Amprius lit-ion battery's kWh is

$$\text{kWh} = \frac{1150 \times 1}{1000}$$

$$1.15 = \frac{1150 \times 1}{1000}$$

So, for every litre in volume of the battery used it can store 1.15 kWh.

Using these numbers it can be calculated that on average, 17L in Amprius battery is required to power a household for a day. This translates to 0.017 m³ of battery. As times of crisis can be unpredictable the generator should be made to provide a generous amount of electricity. 0.5 m³ of battery is equivalent to 500 litres. A generator holding this amount of batteries can produce 575 kWh of electricity, charging an average household for 29 days.” (Liu et al. 2020) (Niu et al. 2024) (Mistry & Mukherjee 2018) (Energy 2023)

“To best optimise the charging of the battery, there must be a balance between the voltage output of the PV panels and the voltage intake of the battery. The average charging efficiency of PV panels is between 21% when tested in standard lab conditions and 10.5% without active cooling. As the product will be operating in real world conditions and currently isn’t designed with a cooling system, it will be assumed the PV panels efficiency is 10.5%.

To find the size of the PV panels required in kW relative to size is:

1. Estimate daily battery charge needed.

Where SOC_{max & min} are the state of charge and the battery capacity is in relation to the battery outlined above. Assume SOC_{max} is 95% as it is unrealistic to assume a charge efficiency of 100% and SOC_{min} is 0.2 as it is assumed that is the lowest percentage of usable battery that should be reach before being discharged.

2. Estimate daily solar output per kW installed.

The solar irradiance of Blacktown is assumed to be similar to Wagga Wagga as they are located close to each other and Wagga Wagga falls under the Sydney city domain. Assume a loss of 15% due to factors such as temperature, shading and inverted losses.

Or assume the common rule of thumb, 5 kWh/day per kW installed

3. Estimate PV size

$$\text{PV size (kW)} = \frac{\text{Battery Charge needed per day (kWh)}}{\text{Average PV output per kW per day (kWh)}}$$

$$\text{PV size (kW)} = \frac{0.8625 \text{ (kWh)}}{5 \text{ (kWh)}}$$

$$0.1725 \text{ kW} = \frac{0.8625 \text{ (kWh)}}{5 \text{ (kWh)}}$$

4. Estimate size in area

$$\text{Area (m}^2\text{)} = \frac{\text{PV size (kW)}}{\text{Efficiency}}$$

$$\text{Area (m}^2\text{)} = \frac{0.1725 \text{ (kW)}}{0.105}$$

$$1.64 \text{ (m}^2\text{)} = \frac{0.1725 \text{ (kW)}}{0.105}$$

So, for every litre of battery size, 1.64 m^2 of PV is required for optimal energy production. This translates to 820 m^2 of solar panels being needed for 0.5 m^3 of batteries, which is unrealistic. A realistic amount of PV panels for the product will be around 0.75 m^2 . While this is far from what is needed for optimal energy production, the product is design to be used in emergencies and not everyday use. This means that it can sit out and charge for the majority of the time, Its large energy storage of 575 kWh also suggests that it is unlikely it will need to be charged during a disaster. Another option can be to add an additional option for a source of power so that the user can charge the product from their normal source of electricity. They can then use the solar panels to refill the battery from any loss that may occur, ensuring it remains at its maximum charge. (Khanal et al. 2024) (Akrouh et al. 2024b) (Joshi et al. 2019)

Size of Generator

“The generator will contain 500 litres of batteries, equivalent to 0.5 m^3 and having 575 kWh. The average PV panel is $2400\text{mm} \times 1800\text{mm} \times 40\text{mm}$, which can be taken to decide the lengths of the generator. $1500\text{mm} \times 1400\text{mm} \times 1000\text{mm}$ will be the lengths of the generator. The total area will be filled 23% by the batteries, 8% by the solar panels and have remaining 69% of room for other components such as the DC/DC controller and microcontroller, as well as wiring and can leave room if a cooling system is to be installed in future iterations. The PV panels will also fold ontop of each other in transit to increase their area as well as to protect. This box will be joined to a flatbed type trailer for easy movement. (see drawing in figure 1 for reference)”

Case Material

“The case material of the generator is essential to keep the interior components protected from the elements. It should be designed to withstand conditions including, fire, floods and extreme heat. An option could be precast concrete with polypropylene fibres, which is generally lighter than traditional concrete, but still weighs between 2251 to 2347 kg/m^3 , which is too heavy to make the product practical for household use. A better option for the case is an Aluminium Composite Panel (ACP) with internal insulation and coating layers. The ideal structure of this is three layers with the outside being the fire rate aluminium composite panel, the middle layer consisting of aerogel or a fire rated foam insulation and the inner casing made from a FR4-grade plastic or aluminium sub-frame. This casing is effective against harsh weathers like fire, as aluminium has a high melting point of 660°C whilst also retaining its structural integrity. Also, if the ACP is made from a high-performer fireproof material it will be able to withstand 1500 degree for around 20-30 minutes. Also, researchers tested the impact resilience of a 400g/m^2 ACP it was able to with stand 11kN this is more than standard flooding in Blacktown that is around $5,000\text{N/m}^2$. When coated or anodised it is also rust resistant and non-porous, while also being able to be fully sealed, preventing water damage on internal components. ACD is easy to bend into size and is lightweight making it more practical to use. It weighs between 5.5 to 7.5 kg/m^2 , as the total surface are as previously worked out is 2.34285m^2 , the total weight of the shell can be estimated to be between 12.89 to 17.57 kg.” (Balasbaneh, Marsono & Gohari 2019) (Amran et al. 2022) (Alshebani & Wedawatta 2014) (Khoury 2000)

Product Diagram

External Product Design

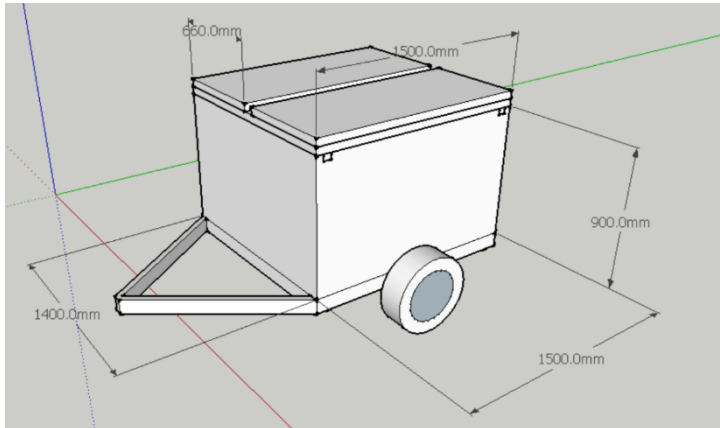


Table 1: Statement of Compliance

Figure 4: Dimensions of Exterior Final Design Front View

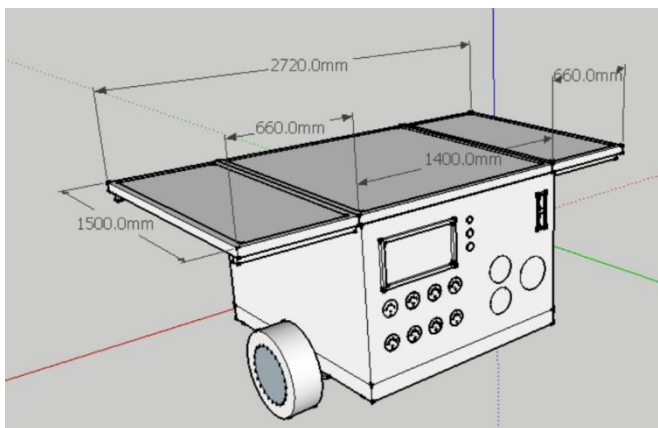


Figure 5: Dimensions for Exterior Final Design Back View with Solar Panels

Also, the group has provided a SketchUp Link that has the Final Design with the correct scales to it:

https://app.sketchup.com/share/tc/australia/2lJr4iEejKc?token=Mjybg6oHdXsGp6237it4i9bD5lwumFpMr4UUe_csbyK2td9Of19D5LyfHH4jqlBL&source=web

Internal Product Design

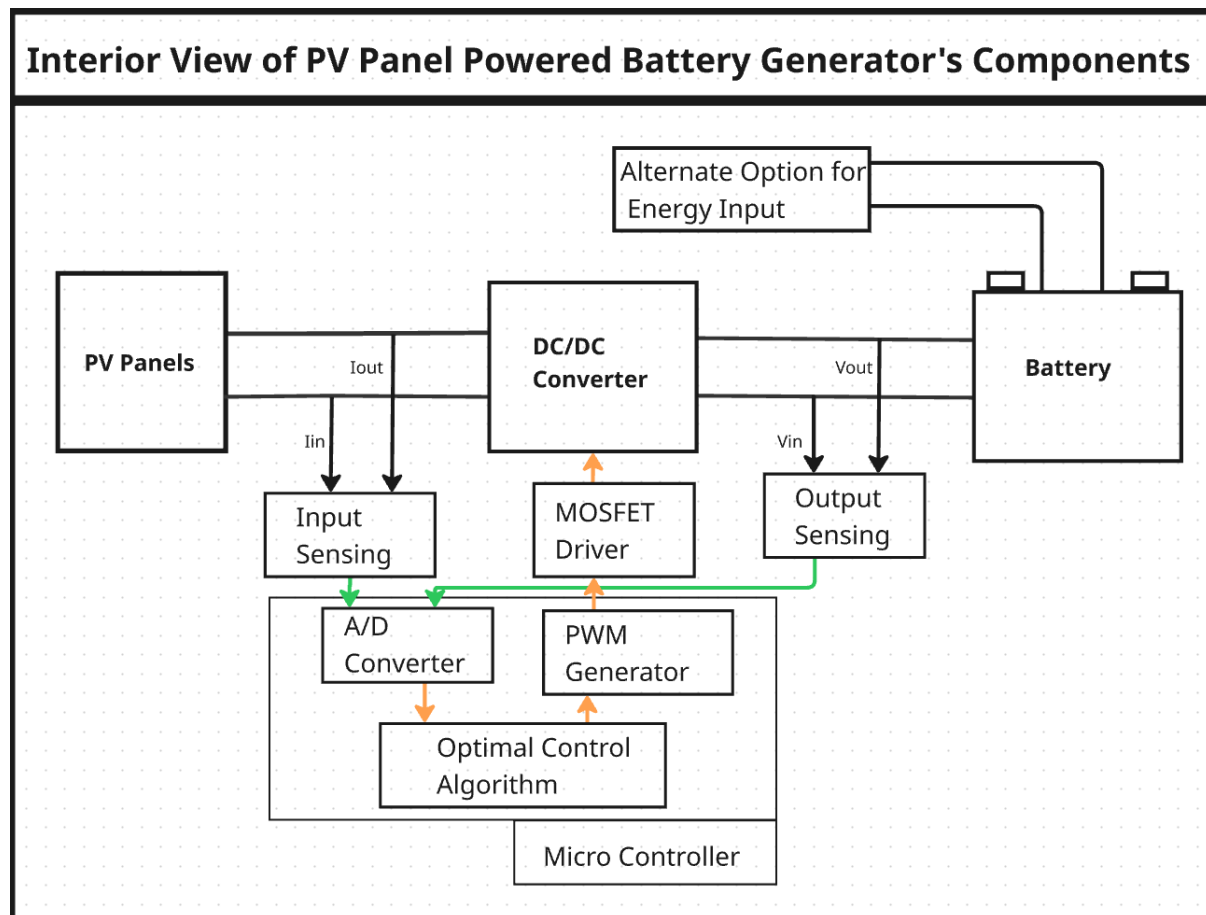


Figure 6: Above is a not to scale design for the interior layout of the product.

Constraint	Compliant	Evidence
<i>Produces enough energy for 2 week (25 kWh per day)</i>	<i>Yes</i>	<i>The generator stores 575 kWh will supply the average Sydney household for close to month. (see Preliminary Design)</i>
<i>Safe around the community (does it follow one of the guidelines)</i>	<i>Yes</i>	<i>Plugs in electrical outputs to prevent any dangerous mishaps from occurring. (see Final Design)</i>
<i>Durable material (can it withstand 5,000N/m² and 600 degrees)</i>	<i>Yes</i>	<i>the case is an Aluminium Composite Panel (ACP) with internal insulation and coating layers (see Preliminary Design)</i>
<i>Environmentally friendly (no more than 0.275 kg CO₂-e/kWh)</i>	<i>Yes</i>	<i>A solar powered generator becomes carbon neutral after 3 years. (see Introduction)</i>
<i>Adaptable for 3 situations</i>	<i>Yes</i>	<i>It can withstand heatwaves, bushfires and flooding. (see Preliminary Design)</i>

6 Discussion

The verified design overview summarises how the current project concept was assessed against its constraints, identifies where requirements were achieved or not and outlines how it will fair in real world implementation.

The verified design addresses the problem statement that is provided at the beginning of the report. It provides a reliable power source for affected community members and addresses the issue of fresh food; however, this may not address the problem of clean water during natural disasters which is hard to control with energy.

A design constraint not compliant with the design is the durability of the material the design is going to use; this is because the team is relaying on online research rather than doing physical tests. Some of the online results aligned with the design's needs however they don't accurately represent the conditioning that Blacktown's extreme weather would cause on the materials. This could lead to a structural failure or an increase in maintenance costs.

A design objective not compliant with the design is the cost-effective this is because the team cannot get a clear estimation for the funds and resources. With digital models alone it doesn't give an accurate prediction on the expenses, as well as the material costs, labor and unseen challenges that could arise during production.

The handover of the verified design may take a year or two as fundraising may need to be considered to limit the amount of cost burden on the council, it will also need a place to be stored which can take time to find a suitable location that won't affect the durability or operation of the generators. The roll out of the generators shouldn't take long as they are mobile and can be transported easily during disasters, once the council have access to them it won't be difficult to use.

What should have been done differently during the early stages of the design process, is focus on the most basic aspects first like what components would be used while also considering the durability and mobility of the generator. This would have been much easier to reach a final design more efficiently and limit the amount of time that has been spent on reaching the verified design.

Additional calculations and work outside of the design process...

6.1 Next Steps / Future Work

The next steps for this design will be to create a pilot design, as only an alpha prototype has been created with the most basic components. This pilot design will contain all components and contain all the materials that the team have researched throughout the screening process and the preliminary design research. To consider cost in more detail is a major next step, more research on making the design more cost effective for the Blacktown Council to implement to ensure that the community don't suffer financially because of the design.

The design will need to be tested and qualified professionals will have to be invited to look at the product and give feedback on areas where the group can improve the design. Possibly implementing pilot programs in other regions to trial its use and to test adequate for the Blacktown council to implement fulltime.

Finally, a community initiative will be created to ensure the public know that these SolarAid generators are available to them when they need them during natural disasters.

6.2 Limitations

Further consideration into the durability and efficiency needed to be used in the design phase, as the efficiency of the generator is still a major concern. This will be included in the next steps to improve the design from the alpha prototype to the pilot design.

7 Conclusions

The proposed design is a solar power generator that will provide a reliable and clean source of energy to the community of Blacktown during recurring flood events and will speed up the road to recovery for the region. It is a portable generator so that it can be given to households that need energy so that they can survive during adverse events. It will be further tested and changes made from the alpha prototype to ensure it meets all specified health and safety codes as well as meeting the expectations of the Blacktown City Council.

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Appendices

Appendix A: Groupwork Contribution Table

Section	Person(s) responsible for writing and percentage	Person(s) responsible for checking / reviewing
Executive Summary	[REDACTED] 100%	[REDACTED], Hugh Williams, [REDACTED] [REDACTED], [REDACTED]
Introduction	Hugh Williams 100%	[REDACTED], [REDACTED], [REDACTED] [REDACTED]
Literature Review	[REDACTED] 50%, [REDACTED] 50%	Hugh Williams, [REDACTED]
Design Requirements	[REDACTED] 100%	[REDACTED], Hugh Williams, [REDACTED] [REDACTED]
Methodology	[REDACTED] 100%	[REDACTED], Hugh Williams, [REDACTED] [REDACTED], [REDACTED]
Results	[REDACTED] 50%, Hugh Williams 50%	[REDACTED], [REDACTED]
Discussion	[REDACTED] 33%, Hugh Williams 33%, [REDACTED] 33%	[REDACTED]
Conclusion	[REDACTED] 100%	Hugh Williams, [REDACTED], [REDACTED] [REDACTED]
Appendices	[REDACTED], Hugh Williams, [REDACTED] [REDACTED], [REDACTED]	

Appendix B: Proof of Team Meetings

21/3 went for 12:30-2:30

We did problem statement, design statement, design constraints and objectives and design insights.

28/3 12:30-2:00

Worked on PowerPoint, Ideation and refining ideas.

4/4 12:30 - 2:00

We worked on the presentation and did the screening to find our final design.

10/4 4:15 – 5:30

Preparing for the presentation and finishing everything up on AT2

11/4 12:30 – 2:00

Start working on AT3 and converting everything from AT2 to AT3

2/5 12:30 - 2:00

Assign task for AT3 for each group member and work on AT3

9/5 12:30 - 2:00

Working through AT3

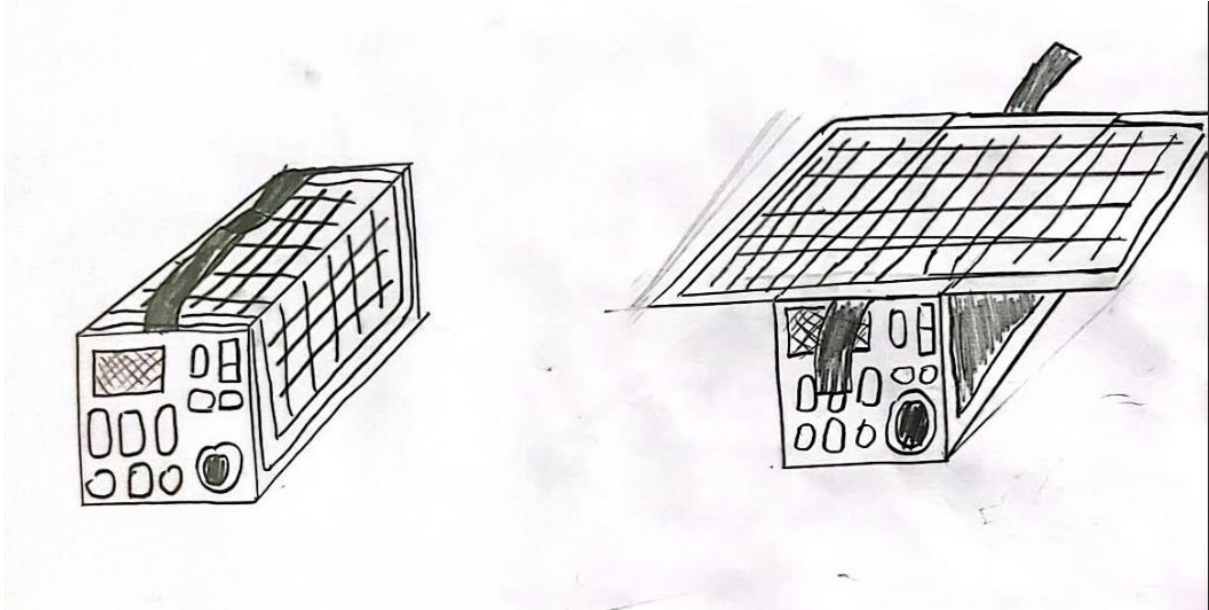
16/5 12:30 - 3:20

Working through AT3 and finishing parts up on it

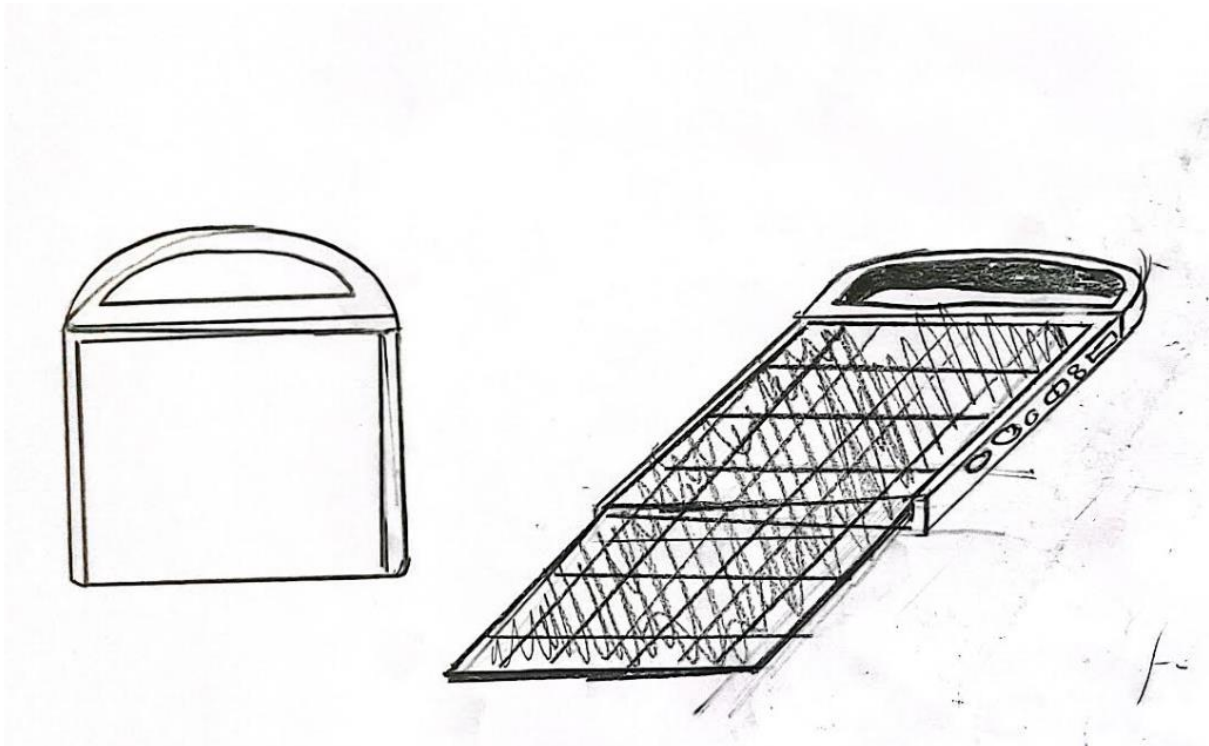
Appendix C: Design Images

External Product Design

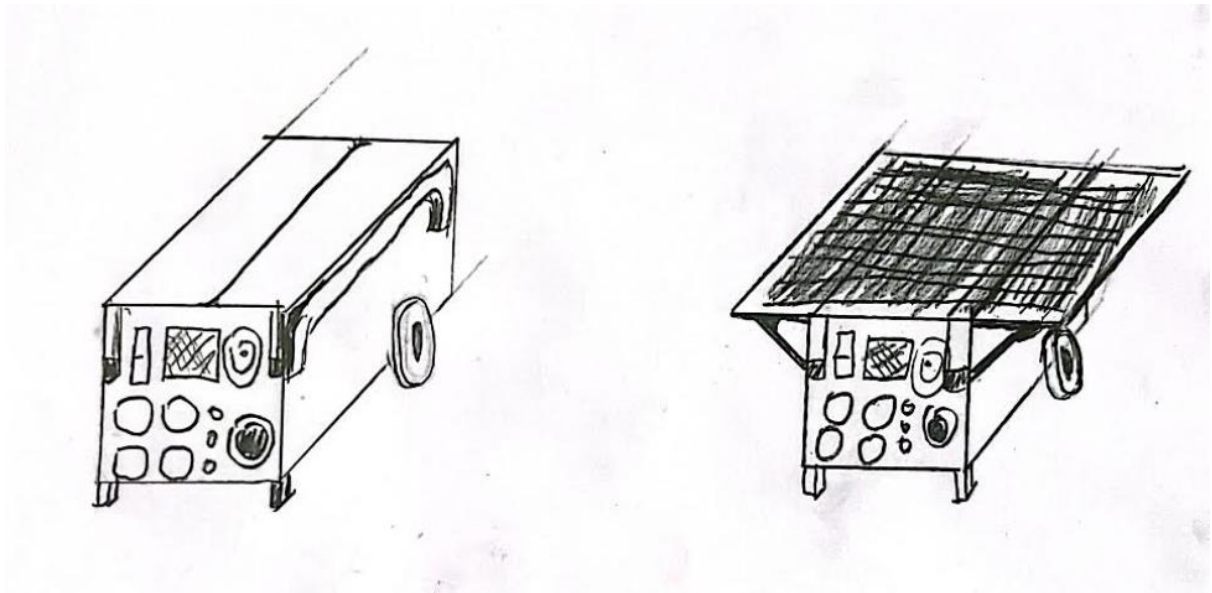
Sketch Designs



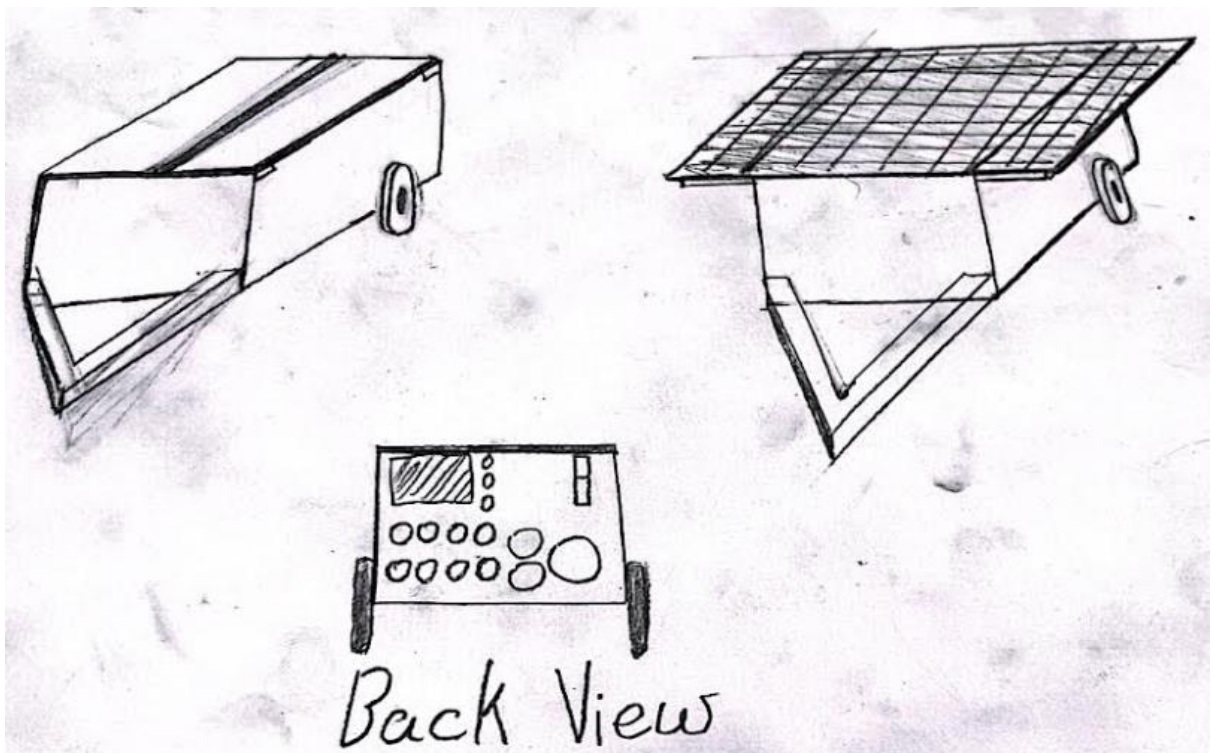
Exterior Design 1



Exterior Design 2

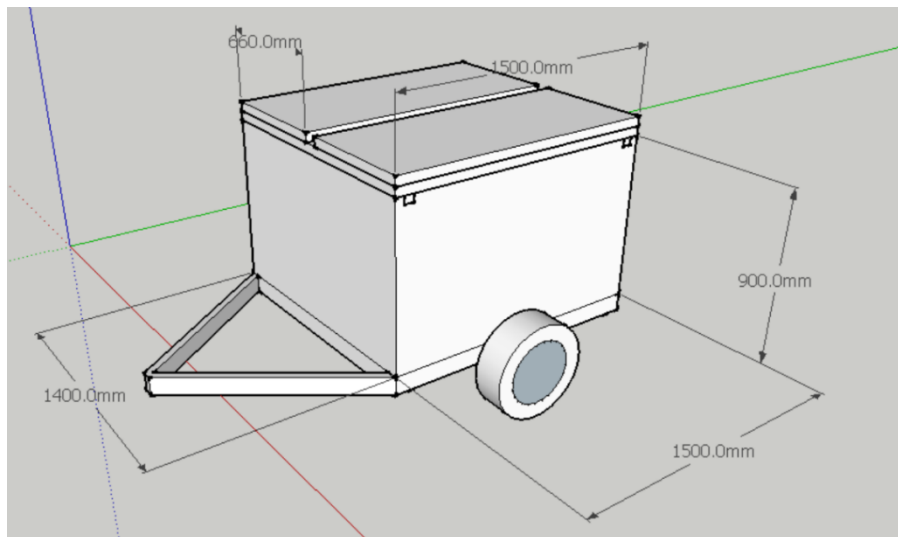


Exterior Design 3

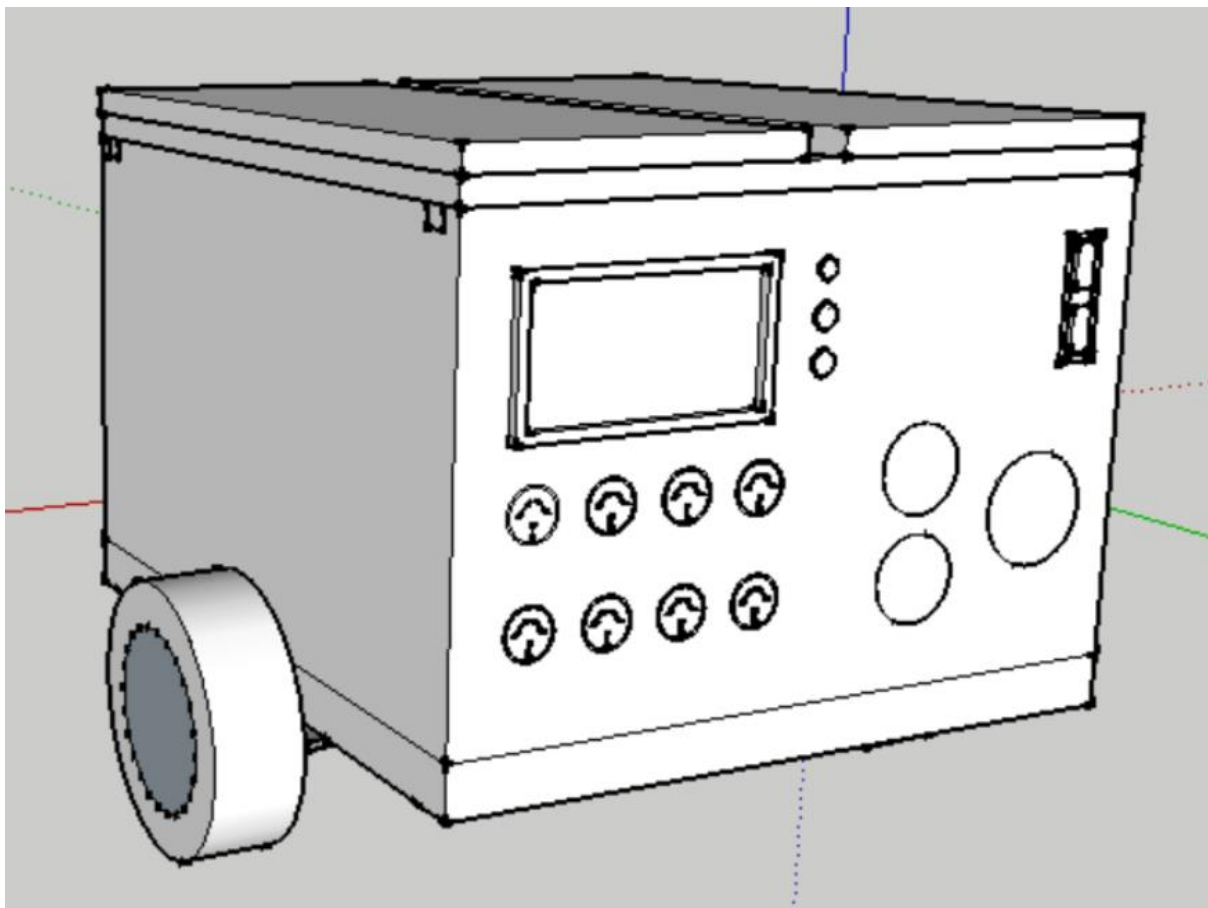


Exterior Design 4

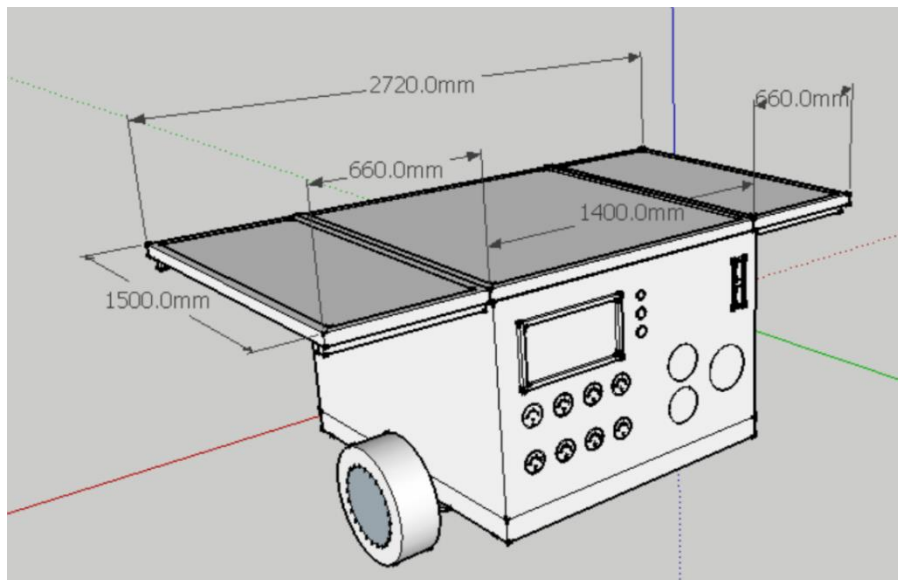
Digital Designs



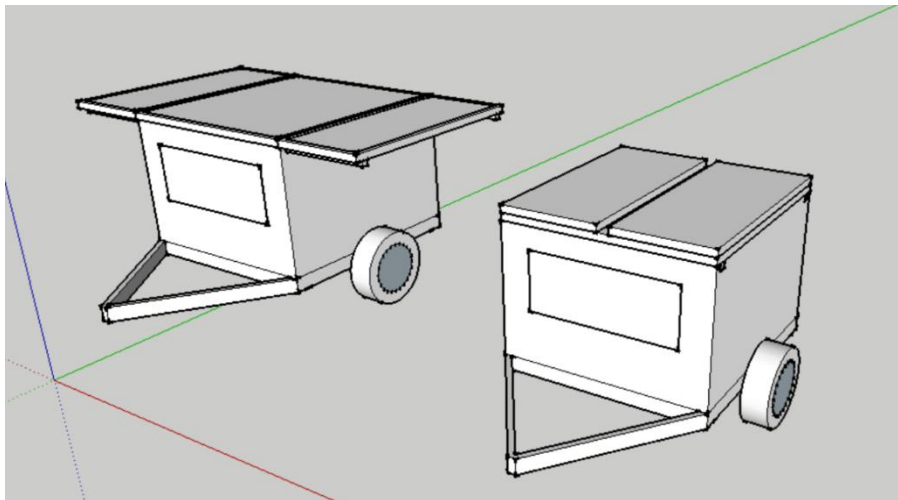
Dimensions of Exterior Final Design Front View



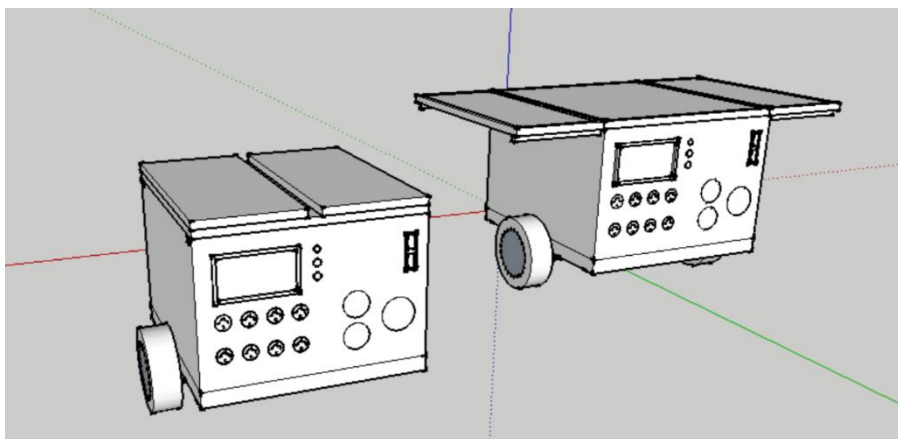
Exterior Final Design Back View



Dimensions for Exterior Final Design Back View with Solar Panels

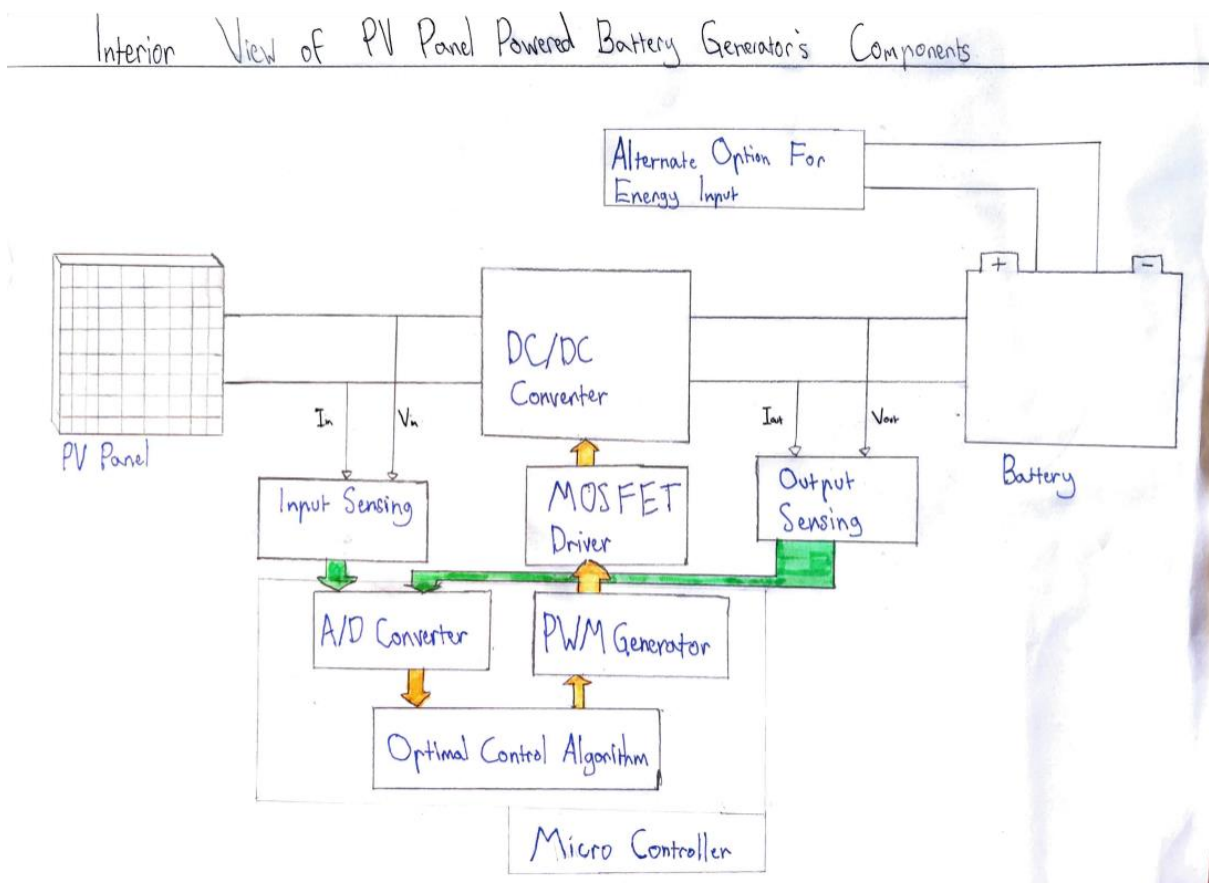
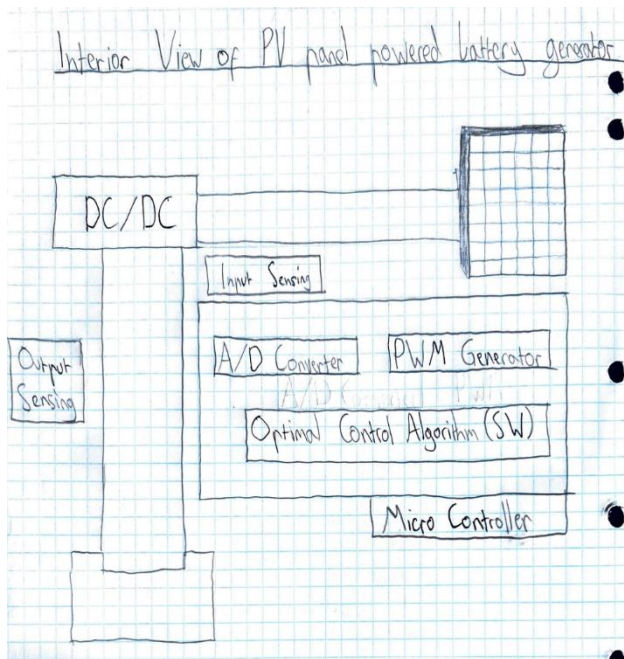


Exterior Final design Front View for Open and Closed Solar Panels



Exterior Final design Back View for Open and Closed Solar Panels

Interior Design Initial



Interior Design Unpolished Final

Interior View of PV Panel Powered Battery Generator's Components

